DAY 15 | PRE-READ

Checking Your Work and Moving Data

A light read before Day 15. About 15 minutes, no code required.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Why read this?

Today has two halves. In the morning you learn to write tests: small programs that check your programs. In the afternoon you step into data engineering: how raw data travels from where it is created to where it helps people decide.

1. Tests are like a pilot's checklist

Before every flight, pilots run through a checklist even though they have flown thousands of times. It takes minutes and prevents disasters. Automated tests are a checklist for code. Every time you change something, the computer re-checks everything in seconds.

Key idea: tests give you the confidence to change code without fear of breaking what already works.

2. Three sizes of test

Test size	Everyday example
Small (unit)	Checking that one light switch works.
Medium (integration)	Checking that the switch, wiring and bulb work together.
Large (end to end)	Walking into the room at night and seeing the light come on.

3. An assistant can draft the checklist NEW (v4)

OpenAI Codex can write a first draft of your tests in seconds. But a checklist is only useful if every line checks the right thing. You still read each test and ask: is this really what the rule says?

4. The journey of data

Think of water. It falls as rain (data is generated by apps and devices), flows into reservoirs (data is collected, or ingested), is stored, then cleaned at a treatment plant (data is processed), and finally arrives at your tap (dashboards, reports and AI models). Data engineers build and maintain the pipes.

5. Clean first, or store first?

ETL cleans data before storing it, like washing vegetables before putting them in the fridge. **ELT** stores everything first and cleans it later inside a powerful database, like putting the shopping away

quickly and preparing it when you cook. Both are valid; modern cloud platforms often prefer ELT.

6. Think about these before class

- What checks do you do before submitting an important document at work?
- Name three places where data about you is generated every day.
- What could go wrong if the same support ticket were loaded into a report twice?

Anshin (安心): peace of mind